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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
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TITLE OF THE PROJECT		
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Abstract

LifeLink – Emergency Blood & Medicine Request Network is a web-based emergency healthcare support system designed to provide a fast, organized, and reliable way for people to find essential medical resources during critical situations. In medical emergencies, patients and their families may face significant difficulties in finding suitable blood donors, locating required medicines, and communicating with available volunteers or pharmacies within a limited amount of time. Traditional approaches often depend on personal contacts, phone calls, social media groups, or visiting multiple hospitals and pharmacies, which can result in delays and difficulties in coordinating emergency assistance. LifeLink aims to address these challenges by providing a centralized digital platform that connects people who need emergency medical assistance with blood donors, volunteers, and medicine providers.

The proposed system allows users to create accounts, securely log in, manage their profiles, and access different emergency healthcare services through a single platform. Users who are willing to donate blood can register as donors by providing information such as blood group, location, contact details, availability, and other relevant information. A person requiring blood can search for available donors based on **blood group and location**, helping them identify potentially suitable donors more efficiently. The system also provides an **Emergency Blood Request** feature through which users can submit important details such as the required blood group, number of units, hospital or treatment location, urgency level, and request status. These requests can be viewed and managed through the platform, allowing available donors and volunteers to respond to emergency situations.

The system also includes a **volunteer management module** that allows individuals to register as volunteers and assist in coordinating emergency requests. Volunteers can view active requests, respond to suitable requests, and help connect patients with available resources. The platform maintains request statuses such as **Open, Accepted, In Progress, and Completed**, which provides better visibility of the emergency-response process and helps users understand the current status of their requests.

In addition to blood donation services, LifeLink provides a **Medicine Finder and Pharmacy Network**. Users can se arch for required medicines and view pharmacies that may have the requested medicine available. Pharmacy information such as pharmacy name, location, contact details, and medicine availability can be maintained in the system. Location-based services can be enhanced using the **Google Maps API**, allowing users to identify nearby donors and pharmacies. This can reduce the time required to search for emergency medical resources and improve the overall usability of the platform.

To make the system more intelligent, LifeLink can incorporate modern AI technologies such as an **AI-based donor recommendation and matching system**. The system can consider factors such as blood group,

donor availability, location, and emergency priority to rank potentially suitable donors. An AI assistant using technologies such as the **Google Gemini API** can also be incorporated to help users navigate the platform and understand the available services. These AI features are intended to support users and improve resource discovery rather than replace professional medical advice or make medical decisions.

The application is developed using modern web technologies including **Java, JSP, HTML5, CSS3, and JavaScript**. **MySQL** is used for storing user, donor, emergency request, volunteer, medicine, and pharmacy information, while **JDBC** provides connectivity between the Java application and the database. **Apache Tomcat** is used to run and deploy the Java web application. **REST APIs** provide a modular mechanism for communication between application components and external services. **Postman** can be used for API testing, while **JUnit** supports automated testing of Java components.

The project follows the principles of **Adaptive Software Engineering and Agile Scrum methodology**. Instead of developing the complete system in a single fixed cycle, the application is divided into multiple iterations or sprints. Requirements can be reviewed after each sprint, and changes can be incorporated based on user feedback and project requirements. **Jira is used to manage the product backlog, user stories, tasks, bugs, and sprint progress**. **Figma can be used to design and modify the user interface before implementation. Git and GitHub provide version control and support collaborative development**. Docker is used to containerize the application and its required services, making the system easier to configure, test, and deploy consistently across different environments.

The development process follows a continuous cycle of **planning, development, testing, feedback, adaptation, and improvement**. For example, an initial blood donor search feature can be developed in one sprint, evaluated through testing and user feedback, and then improved by adding location filtering, map integration, or intelligent donor recommendations in subsequent iterations. This adaptive approach allows the system to respond to changing requirements and provides flexibility throughout the development process.

The primary objective of LifeLink is to **reduce the time and effort required to locate emergency blood and medicine resources** by providing multiple services through one centralized platform. By integrating blood donor management, emergency request handling, volunteer coordination, medicine search, pharmacy information, location-based services, AI-assisted recommendations, and modern software development practices, LifeLink demonstrates how web technologies and adaptive development methodologies can be combined to create a practical emergency healthcare support system. The project can be further enhanced in the future with real-time notifications, mobile applications, hospital and blood-bank integration, advanced AI recommendation models, real-time location tracking, and secure authentication mechanisms.

Keywords: LifeLink, Emergency Healthcare, Blood Donor Management, Emergency Blood Request, Medicine Finder, Volunteer Network, REST API, MySQL, Java, Docker, Agile Scrum, Adaptive Software Engineering, Artificial Intelligence, Google Maps API, Jira.

Signature of the Guide HOD